Youqi Yang

PhD Candidate · Department of Biostatistics · University of Michigan

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Education

University of Michigan *Ann Arbor, MI*
Doctor of Philosophy in Biostatistics *August 2023 - present*
- Advisors: Dr. Bhramar Mukherjee and Dr. Walter Dempsey

University of Michigan *Ann Arbor, MI*
Master of Science in Biostatistics *August 2021 - April 2023*
- Advisor: Dr. Thomas Braun

Tsinghua University *Beijing, China*
Bachelor of Science in Biological Science *August 2017 – June 2021*
- Advisor: Dr. Xun Lan

University of California, Berkeley *Berkeley, CA*
Summer Session *July 2019 – August 2019*

Professional Experience

Graduate Student Research Assistant *Ann Arbor, MI*
Data Science for Dynamic Intervention Decision Making Center *August 2025 - present*
- Advisor: Dr. Walter Dempsey

Graduate Student Research Assistant *Ann Arbor, MI*
Department of Biostatistics at University of Michigan *May 2022 - August 2025*
- Advisor: Dr. Bhramar Mukherjee

Research Assistant *Ann Arbor, MI*
Department of Computational Medicine and Bioinformatics at University of Michigan *May 2022 - August 2022*
- Advisor: Dr. Lana Garmire

Undergraduate Research Assistant *Beijing, China*
Department of Basic Medical Sciences at Tsinghua University *August 2019 – June 2021*
- Advisor: Dr. Xun Lan

Publications

Methodological

Yang, Y., Nahum-Shani, I., Ertefaie, A., and Dempsey, W. (2026). Causal mediation analysis of interventional effects on binary proximal outcomes in micro-randomized trials. *Manuscript*. `https://www.dropbox.com/scl/fo/1zoumgv2grenvn6mc512r/ALyCkMJQ_XBrZvgTRb192H0?rlkey=tf3i8eglfmu5qnwcqsxvz0rvj&st=rnyn4l1j&dl=0`

Yang, Y., Dempsey, W., and Mukherjee, B. (2026). Data integration for estimating subgroup-specific conditional average treatment effects (CATEs) using coarsened external information in randomized trials. *arXiv preprint arXiv:2604.20667*. [Invited revision at *Biometrics*.]

Unjitwattana, T., Huang, Q., Yang, Y., Tao, L., **Yang, Y.**, Zhou, M., Du, Y. and Garmire, L.X. (2025). Single-cell RNA-seq data have prevalent blood contamination but can be rescued by Originator, a computational tool separating single-cell RNA-seq

by genetic and contextual information. *Genome Biology, 26(1)*, 1-13.

Yang, Y., Dempsey, W., Han, P., Deshmukh, Y., Richardson, S., Tom, B., and Mukherjee, B. (2024). Exploring the big data paradox for various estimands using vaccination data from the global COVID-19 Trends and Impact Survey (CTIS). *Science Advances, 10*, eadj0266. https://doi.org/10.1126/sciadv.adj0266

Applied

Admon, A. J., **Yang, Y.**, Lee, K. T., Nagle, M. T., Kamphuis, L., Caldwell, E., Hope, A. A., Eaton, T. L., Jolley, S. E., Peltan, I. D., Iwashyna, T. J., Baron, R. M., Friese, C. R., Mukherjee, B., and Hough, C. L. (2026). Symptom trajectories in the year after COVID-19 hospitalization: A multicenter prospective cohort study. *CHEST Pulmonary*, 100276.

Yang, Y., Zimmermann, L., Pramanik, S., Wahl, B., and Mukherjee, B. (2025). The role of financial stress, food insecurity, and COVID-19-related illness concerns shaping mental health in five South Asian countries (2020–2022). *PLOS Global Public Health, 5*(8), e0004704.

Yang, Y., Sun, A., Zimmermann, L., and Mukherjee, B. (2024). Impact of pandemic-related worries on mental health in India from 2020 to 2022. *npj Mental Health Research, 3*, 57. https://doi.org/10.1038/s44184-024-00101-x.

Grieve, R., **Yang, Y.**, Abbott, S., Babu, G. R., Bhattacharyya, M., Dean, N., Evans, S., Jewell, N., Langan, S. M., Lee, W., Molenberghs, G., Smeeth, L., Williamson, E., and Mukherjee, B. (2023). The importance of investing in data, models, experiments, team science, and public trust to help policymakers prepare for the next pandemic. *PLOS Global Public Health, 3*(11), e0002601. https://doi.org/10.1371/journal.pgph.0002601

Awards, Fellowships, & Grants

2026	**Candidate Rackham Graduate Student Research Grant**, Rackham Graduate School at University of Michigan	*$3,000*
2026	**Rackham Conference Travel Grant**, Rackham Graduate School at University of Michigan	*$900*
2025	**Rackham Conference Travel Grant**, Rackham Graduate School at University of Michigan	*$900*
2024	**Rackham Conference Travel Grant**, Rackham Graduate School at University of Michigan	*$900*
2020	**Integrated Excellence Award**, School of Life Sciences at Tsinghua University	
	Excellent Academic Performance, School of Life Sciences at Tsinghua University	

Presentations

Invited Presentations

Yang, Y., Nahum-Shani, I., Ertefaie, A., and Dempsey, W. (2026). Causal mediation analysis of interventional effects on binary proximal outcomes in micro-randomized trials. *Causal AI for Decision Making Workshop, University of Michigan*, Ann Arbor, MI.

Yang, Y., Dziak, J., Nahum-Shani, I., Ertefaie, A., and Dempsey W. (2026). Causal mediation analysis of interventional effects on binary proximal outcomes in micro-randomized trials. *Conference on Intervention Optimization*, Rockville, MD.

Yang, Y. (2025). Exploring the data from the global COVID-19 Trends and Impact Survey (CTIS) in India. *Survey Methodology Lab*, Ann Arbor, MI.

Yang, Y. and Mukherjee, B. (2024). The big data paradox and mental health trends using COVID-19 Trends and Impact Survey (CTIS) data. *Center for Global Health Equity at University of Michigan India Platform Meeting*, Ann Arbor, MI.

Contributed Presentations

Yang, Y., Dempsey, W., and Mukherjee, B. (2026). Data integration for estimating subgroup-specific conditional average treatment effect (CATE) using coarsened external information in randomized trials. *Joint Statistical Meeting*, Boston, MA.

Yang, Y., Dempsey, W., and Mukherjee, B. (2026). Data integration for estimating subgroup-specific conditional average treatment effect (CATE) using coarsened external information in randomized trials. *Eastern North American Region Spring Meeting*, Indianapolis, IN.

Yang, Y., Dempsey, W., and Mukherjee, B. (2025). Comparing methods for estimating treatment heterogeneity in a randomized controlled trial with external marginalized estimates. *Eastern North American Region Spring Meeting*, New Orleans, LA.

Yang, Y., Dempsey, W., Han, P., Deshmukh, Y., Richardson, S., Tom, B., and Mukherjee, B. (2024). Exploring the big data paradox for various estimands using vaccination data from the global COVID-19 Trends and Impact Survey (CTIS). *Joint Statistical Meeting*, Portland, OR.

Yang, Y., Dempsey, W., Han, P., Deshmukh, Y., Richardson, S., Tom, B., and Mukherjee, B. (2024). Exploring the big data paradox for various estimands using vaccination data from the global COVID-19 Trends and Impact Survey (CTIS). *Eastern North American Region Spring Meeting*, Baltimore, MD.

Yang, Y., Dempsey, W., Han, P., and Mukherjee, B. (2023). What can we learn from observational data wrinkled with selection bias? A case study using Facebook COVID-19 Trends and Impact Survey in India and the US. *Michigan Student Symposium for Interdisciplinary Statistical Sciences*, Ann Arbor, MI.

Yang, Y., Huang, Q., Zhou, M., Benny, P., Schlueter, R., Lassiter, C., and Garmire, L. (2022). A novel single-cell RNA-seq analysis workflow for the human placenta. *Society for the Study of Reproduction Annual Meeting*, Spokane, WA.

Teaching Experience

2023 **Student Coordinator**, Big Data Summer Institute *Ann Arbor, MI*

Professional Development

Service

2023+ **Biostatistics Graduate Student Working Group**, President *Ann Arbor, MI*
2024-2025 **Biostatistics PhD Admissions Committee**, Student Member *Ann Arbor, MI*
2023-2024 **Statistics in the Community**, Committee Member *Ann Arbor, MI*
2021-2024 **Biostatistics Diversity, Equity & Inclusion (DEI) Committee**, Student Representative *Ann Arbor, MI*
2020-2021 **Qingxin Times**, Managing Editor *Beijing, China*

Peer Review

BMC Public Health
Harvard Data Science Review
Health Science Reports
Health Services and Outcomes Research Methodology
Population Health Metrics
Scientific Data

Organization

American Statistical Association
International Biometric Society